Chapter 1

Mass, count, and in-between: Insights from Asante Twi

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Asante Twi is a language where numerals and mass nouns may freely combine without the need for a classifier or container phrase. Other languages with this feature, such as Yudja and Nez Perce treat every noun as grammatically count and are considered to be the same type according to Chierchia's 2021 typology of mass-count systems. Against the predictions of this typology, Asante Twi has nouns that are notionally count and grammatically mass, called object-mass nouns. These nouns are likely a result of the decaying noun class system. Furthermore, the mass-count system of Asante Twi more closely resembles English, despite the difference in how the two languages treat numeral-noun constructions. Ultimately, the whole mass-count system should be considered to make predictions on the existence of object-mass nouns in a language.

1 Introduction

Chierchia (2021) has laid out three types of languages with respect to how numerals combine with nouns: Type I Languages include languages like English, French, and German where numerals combine directly with some nouns, but require a classifier or container phrase to combine with others, with this distinction aligning closely with the mass-count distinction in these languages.

(1) Type I: English
   a. two birds
   b. two piles of flour
   c. *two flours

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Type II languages are those such as Mandarin (Zhang 2007), Japanese (Iida 2021), and Bangla (Dayal 2012). These languages require classifiers for numerals to combine with any noun.

(2) Type II: Mandarin

a. wû gè jīdàn
five CL egg
'five eggs' (Bale & Gillon 2020: ex. 5a)

b. sān bēi yóu
three CL oil
'three cups of oil' (Bale & Gillon 2020: ex. 7a)

Lastly, Type III languages allow for numerals to combine freely with any noun without a classifier or container phrase. This type includes languages such as Yudja (Lima 2014) and Nez Perce (Deal 2017). In the coming sections, I will show that Asante Twi behaves like a Type III language.

(3) Type III: Nez Perce

a. kii lepit cickan
DEM two blanket
'these two blankets' (Deal 2017: ex. 22b)

b. lepit kike't hi-sew-n-e
two blood 3SUBJ-fall-P-REM.PST
'Two drops of blood fell.' (Deal 2017: ex. 28)

Cognitive psychologists such as Soja et al. (1991) and Carey & Spelke (1996) have established that children as young as two years old distinguish between concepts such as object and substance before acquisition of count and mass syntax. Even though the three types of languages treat numeral-noun combinations differently, speakers of all three types still cognitively distinguish object and substance terms. Because of this, the mass-count distinction can be thought of in two different ways — cognitively and grammatically. This can be seen in how languages perform quantity comparisons. Barner & Snedeker (2005) showed that certain nouns, but not others in English can be cognitively individuated. Their experiment presented participants with an images of two characters, with one character having one large object, e.g., a shoe or a pile of toothpaste. The other character had three smaller objects or piles. The participant was then asked to compare the quantities that the characters had by asking them which character had more. When presented with shoes, participants overwhelmingly selected the character having three small shoes as having more, and when presented with piles of toothpaste, participants selected the character with the large pile as having more. This shows that objects, such as shoes, are individuated, and compared on a cardinal scale. Conversely, substances, like toothpaste, are not individuated and compared volumetrically. Barner & Snedeker (2005) found that even though words such as *silverware* and *jewelry* are used with mass syntax, they are still cognitively individuated, as participants selected the character having three small spoons as the one having more silverware over the character with one big spoon. Versions of Barner & Snedeker's 2005 experiment have been repeated for Type II languages, such as Japanese (Inagaki & Barner 2016), as well as Type III languages (Lima 2014, Deal 2017). Each experiment came to similar conclusions, that even for languages without a grammatical distinction between count and mass, speakers of such languages still cognitively individuate objects when making quantity comparisons. For Type II and Type III languages, the mass-count distinction seems to exist only cognitively, as substance and object concepts are treated, for the most part, the same by the grammar. Type I languages, however, have a grammatical contrast in how they treat the terms referring to the cognitive concepts of substances and objects.

In English, a Type I language, count nouns can freely combine with numerals and exhibit a morphological singular/plural distinction, shown in (4a). Mass nouns, as in (4b) require a classifier or container phrase to be able to combine with numerals and cannot be pluralized to arrive at a cumulative interpretation.

(4) Combination with numerals, Pluralization

a. **Count**
   one eye, two eyes

b. **Mass**
   one drop of blood, two drops of blood
   #one blood, #two bloods

However, some mass nouns in English can directly combine with numerals and be pluralized, but these are typically restricted to HoReCa (hotels, restaurants, cafes) contexts, and have an interpretation of *n quantities of the standard container of N* or *n quantities of various kinds of N* due to the availability of standard containers of these items, particularly when served in locations such as hotels, restaurants, and cafes (Rothstein 2021). For example, *three wines* can mean 'three kinds of wine' or 'three glasses of wine', but never 'three drops of wine'.

Count nouns and mass nouns also differ in the quantifiers that they can combine with. Quantifiers such as *a few* and *many* are only felicitous with count nouns, while quantifiers like *a bit of* and *much* can only be combined with mass nouns.

(5) Combination with quantifiers

a. **Count**
   a few books, many pens
   #a bit of books, #too much pens

b. **Mass**
   #a few oil, #many dust
   a bit of oil, too much dust

It is possible, however, for certain terms to exhibit a mismatch where concepts that are cognitively understood as objects are treated grammatically as mass. Chierchia (2021) reports that these nouns pattern with mass syntax except for tasks that either do not or only tangentially involve language, such as quantity comparison. In English, this includes terms like *furniture*, *jewelry*, and *mail*. These nouns have been referred to as a variety of names in the literature, such as: 'fake mass nouns' (Chierchia 2010), '*furniture*-like nouns' (Bale 2021), and 'object-mass nouns' (Barner & Snedeker 2005, Erbach & Schoenfeld 2022). For this paper, the term object-mass nouns will be used.

(6) Object-mass nouns

a. Combination with numerals:
   *three furnitures

b. Pluralization:
   *footwears

c. Combination with count quantifiers:
   *a few mails

d. Combination with mass quantifiers:
   too much jewelry

Object-mass nouns, like *footwear* cannot combine with numerals without a classifier (6a). They also cannot take on morphological pluralization to mean

’a cumulation of *x*’ (6b). Additionally, These nouns can only combine with quantifiers reserved for mass nouns, as illustrated in (6c) and (6d). Table 1 provides a summary of the distribution of the various types of nouns in English.

Table 1: Syntactic distribution of mass and count terms in English (Deal 2017: p. 9)

	Count	Object-mass	Mass
Direct combination with numerals	✓	X	X
Pluralization	✓	X	X
Combination with quantifiers *many*, *fewer*	✓	X	X
Combination with quantifiers *much*, *less*	X	✓	✓

The specific vocabulary items that fall into the designation of object-mass nouns is unique to each language. For example, French *meuble* ‘furniture’ behaves as a typical count noun in French (7), but English *furniture* is even the namesake in some of the literature for this class of cognitively count but grammatically mass nouns.

(7) J’ai acheté trois meubles.
I.have buy.PST three furniture
‘I bought three pieces of furniture.’

Reportedly, only Type I languages are capable of having these sort of nouns according to Chierchia (2021). Type II languages, also called classifier languages, have sets of classifiers only for objects. Some classifiers are also specific to certain domains. An object-mass noun in a classifier language would be a noun that is cognitively an object, but cannot combine with classifiers that are grammatically restricted combining with objects. Nouns exhibiting this pattern are unattested according to Cheng, Sybesma, et al. (1998). Furthermore, Chierchia (2021) states that for Type III languages, the lack of furniture-nouns is a result of numeral-noun combinations having an interpretation of *n natural units of N* for count nouns and *n natural quantities of N* for mass nouns. A object-mass noun in these languages would be a count term that is interpreted the same as a mass term when combining with numerals. For example, a potential object-mass noun, such as *rug* would be interpreted as *three aggregates of rug* or *three pieces of a rug*. Chierchia states that this phenomena is so far unattested.

Asante Twi, a Kwa (Niger-Congo) language spoken in Ghana, is shown to be a counter example to Chierchia's 2021 typology. It is a Type III, as numerals in Asante Twi can combine with any noun, but the language does exhibit a class of nouns which are cognitively count but pattern grammatically with mass nouns. This research explores the distribution and source of object-mass nouns in Asante Twi. I argue that Asante Twi shares more in common with Type I languages and that object-mass nouns are born from the now-defunct noun class system. I also provide an analysis for Asante Twi's numeral-noun constructions. The following section §2 discusses the mass-count distinction in Asante Twi. This is then followed by Section §3, a comparison of Asante Twi's system with that of a Type I language, namely English. Section §4 outlines how Asante Twi compares to other Type III languages and Section §5 proposes a null classifier to account for numeral-noun combinations in Asante Twi. Section §6 briefly concludes the paper.

2 Asante Twi

Asante Twi is one of many varieties under the umbrella term 'Akan languages'; about half of all Ghanaians speak a variety of Akan as either a first or second language, with Asante Twi being the most popularly spoken. *Akan* and *Twi* are occasionally used interchangeably in the literature to describe the subject language of this paper. The term 'Akan' most often refers to the languages spoken by the groups of culturally similar people residing between the Volta and Bandama rivers who distinguish themselves from other ethnic groups (Dolphyne 1986)[1].

The data for this research was collected during fieldwork undertaken in September and October 2024 at the University of Ghana, in Legon, Ghana. Seven native speakers (women = 2, men = 5; age: 22-26; average: 23.4) of Asante Twi were consulted, each of whom had grown up in Kumasi, Ghana, and moved to the Greater Accra Region for university studies. The parents of all speakers were also native speakers of Asante Twi.

[1]For a more in depth discussion surrounding the name of the subject language in academic literature, see (Kiyaga-Mulindwa 1980, Dolphyne 1986).

2.1 Asante Twi as a Type III language

All nouns in Asante Twi freely combine with numerals and do not need classifiers or container phrases to get a unit reading, shown in (8)[2].

(8) a. mé.kú-ú n-kótérɛ nan.
I.kill-PST PL-lizard four
'I killed four lizards.'

b. mé.tɔ-ɔ aŋwá mmíénú.
I.buy-PST oil two
'I bought two containers of oil.'

When mass nouns combine with a numeral, they are interpreted as *n contextually salient quantities of N*. To illustrate, (8b) shows that *aŋwá mmíénú* is interpreted as 'two containers of oil', as oil is typically bought in large containers, but *aŋwá mmíénú* could also be interpreted as 'two drops of oil' if permitted by the context, such as if some oil was spilled when cooking.

While Asante Twi treats treats object and substance terms the same when combining them with numerals, the mass-count distinction can be seen in other grammatical constructions in the language.

2.2 The mass-count distinction

The first difference between mass and count nouns in Asante Twi is the availability of a noun-class prefix alternation. For count nouns, singular nouns are marked with [e-/ɛ-], [o-/ɔ-] (modulo vowel harmony), or [a-], while plurals are marked with either [a-] or a nasal[3]. The prefix of singular nouns is occasionally realized as a phonologically null element (∅) due to the attrition of the noun class system. However, the singular-plural distinction in these nouns can be seen nonetheless, as shown by (9b).

(9) **Count nouns**

a. a-kónyá n-kónyá
SG-chair PL-chair

[2] Asante Twi examples are presented in the orthography as defined in Osam (1994). Additionally, high tones are marked and low tones are left unmarked.

[3] For singular nouns that are marked with [a-], their plural counterparts are marked with [N-].

b. ∅-kótérɛ n-kótérɛ
SG-lizard PL-lizard

This contrasts with the mass nouns, such as those in (10), which lack a morphological number distinction, as no other prefix is compatible with these words[4]. Interestingly, the majority of mass nouns have the same prefixes as plural nouns; they are marked with either an [a-] or a nasal.

(10) **Mass nouns**

a. m-frámá / *a-frama
NC-air PL-air

b. a-ŋwá / *ŋ-ŋwá
NC-oil PL-oil

For the most part, the distinction of having or lacking a noun class prefix alternation follows the cognitive distinction of object and substance, therefore showing a grammatical mass-count distinction. This distinction is seen in number marking, as in (9) and (10), and it can also be seen in other constructions, which are discussed below. A morphological number distinction can be considered to be 'count syntax', since it is associated with object nouns, while lacking this distinction can be considered 'mass syntax', as this is associated with substance nouns. However, there does exist a subset of object nouns which align with mass syntax, suggesting the existence of object-mass nouns in Asante Twi. These nouns are cognitively objects, as they have discreet minimal parts that do not overlap, but, like mass nouns, they lack a noun class prefix alternation and the prefix is typically an [a-] or a nasal (Osam 1993)[5].

(11) **Object-mass nouns**

a. á-kútúó / *n-kútúó
NC-orange PL-orange

[4]Where a morphological number distinction exists, prefixes are glossed with SG or PL to reflect that distinction. Words lacking in a morphological number distinction are glossed with NC for Noun Class. No numbering system is used as the noun class is defunct.

[5]While it seems that most nouns lacking a morphological number distinction, some nouns, however, seem to have undergone a rebracketing, where the former class prefix is analyzed as part of the root or was dropped for both singular and plural counterparts. These nouns which don't have a singular-plural distinction, but also don't pattern with mass syntax like *mpoma* 'mountain' or *adwene* 'idea' serve as examples of nouns lacking a singular-plural distinction, but are not object-mass nouns. These nouns are simply number-neutral.

b. n-homá / *a-homá
NC-book PL-book

This group of nouns contains mostly fruits and vegetables, but also some artifacts, animals, or body parts that usually come in multiples or pairs, i.e., *mpomá* 'window(s)', *nwa* 'snail(s)', *nnán* 'leg(s)'.

Aside from their number morphology aligning with mass nouns, object-mass nouns can also combine with mass quantifiers such as *dódó* 'much' and *kakra* 'a bit of' (12b), (12c), which is not possible for count nouns (12a).

(12) a. **Count nouns**
Afia tɔ-ɔ a-/n-konyá *dódó/*kakra
Afia buy-PST SG-/PL-chair a.lot.of/a.bit.of
'Afia bought *a lot of / *a bit of chair/s'

b. **Mass nouns**
mé-nom n-suó dódó/kakra
1SG-drink NC-water a.lot.of/a.bit.of
'I drank a lot of / a bit of water'

c. **Object-mass nouns**
o.di-i á-kútúó dódó/kakra
3SG-eat-PST NC-orange much/a.bit.of
'She ate a lot of / a bit of orange(s)'

Table 2 serves to summarize the distributions of each of the three types of nouns: They are all able to combine directly with numerals. Object-mass nouns and mass nouns both lack a grammatical number distinction, while count nouns have a noun class prefix alternation to express a distinction between singular and plural. Count nouns, however, cannot combine with the quantifiers *dódó* 'much' and *kakra* 'a bit of', where mass and object-mass nouns can. Interestingly, while other quantifiers in the language do exist, Asante Twi does not have quantifiers that combine with only count nouns. I.e., other quantifiers such as *pii* and *bebree*, both meaning 'many' or 'a lot of', felicitously combine with both mass and count nouns.

(13) a. me-hunu-u n-kótérɛ píí / bébréé.
1SG-see-PST PL-lizard many a.lot.of
'I saw many lizards.'

b. ɔ-dɛ enam píí / bébréé ma a-kwada no.
3SG-give meat many a.lot.of to SG DEF
He gave a lot of meat to the child.

Table 2: Syntactic distribution of count, mass, and object-mass nouns

	Count	Object-mass	Mass
	nkonyá ‘chairs’	ákútúó ‘orange(s)’	aŋwá ‘oil’
Direct combination with numerals	✓	✓	✓
Morphological number distinction	✓	X	X
Combination with *dódó* and *kakra*	X	✓	✓

3 Comparing object-mass nouns in English and Asante Twi

Similarities between object-mass nouns in English and Asante Twi can clearly be seen in their syntactic distribution, as well as their semantic denotations. They seem to be the same phenomenon, despite the claim that a Type III language can reportedly not have object-mass nouns. However, differences between the two are seen in the respective semantic domains that the object-mass nouns occur in the two languages, and in their respective diachronic development as a category of nouns.

3.1 Denotation

Building on previous analysis of object-mass denotations, Bale & Gillon (2020) provide a discussion of atomic and non-atomic denotations. They state that an atomic noun such as *shoe* can only denote an entity that is a proper shoe, but not of a part of a shoe nor a collection of shoes. Non-atomic nouns like *oil* then denote any arbitrary observable portion of oil. By this definition of atomic and non-atomic, Bale & Gillon (2020) argue that object-mass nouns are indeed atomic nouns, and include atoms in their semantic denotations. The semantic distinction of atomic and non-atomic therefore aligns the cognitive distinction between substance and object, both of which differ from the syntactic mass-count distinction.

This analysis is furthered by the fact that object-mass nouns and count nouns can be modified by stubbornly distributive predicates, while mass nouns cannot as in (14) (Schwarzschild 2011).

(14) a. **Count**
large cup, square plate

b. **Mass**
*large blood, *square flour

c. **Object-mass**
large weaponry, square furniture

Furthermore, atomic denotations allow for quantity comparison on a cardinal scale, while non-atomic denotations do not. (15) is true in a situation where Lieke has six each of small forks, spoons, and knives, even if Prudence has two each of large forks, spoons, and knives and the weight of all of Prudence's silverware combined exceeds the weight of Lieke's silverware. (15) is evaluated based on the total number of silverware, rather than the total volume of silverware.

(15) Lieke has more silverware than Prudence.

Therefore, in terms of denotations, following (Link et al. 1983, Chierchia 1998, Barner & Snedeker 2005, a.o.) count and object-mass nouns have atomic denotations, while mass nouns have non-atomic ones. Given that the semantic denotation of a noun matches our cognitive conceptualization of it, one can imagine that the denotation for object-mass nouns in Asante Twi are no different. In some recent experimental evidence on Akan, a group of varieties to which Asante Twi belongs, Ladore (2026) shows that plural count nouns are inclusive in downward entailing contexts. In other words, the plural *mmarima* 'men' in the positive sentence, (16a), can be interpreted as 'more than one man'. However, the negative sentence (16b) is interpreted as 'Kojo did not see *one or more men* .

(16) a. Kojo hunu-u m-marima
Kojo see-PST PL-man
'Kojo saw men.'

b. Kojo an-hu m-marima
Kojo NEG-see PL-man
'Kojo didn't see men.'

This means that the denotation for count nouns is different for singular and plural nouns. Singular count nouns denote simply $\{a, b, c\}$, and plural count nouns, in light of Ladore (2026), denote the full semilattice $\{a, b, c, a{\oplus}b, a{\oplus}c, b{\oplus}c, a{\oplus}b{\oplus}c\}$, as shown in Figure 1.

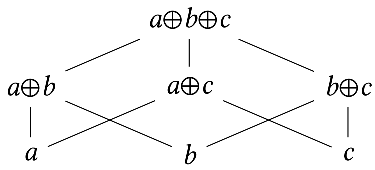


Figure 1: Denotation of plural count nouns

Mass nouns, being non-atomic, then denote the full semilattice, though without atoms included in the denotation, represented by the lack of minimal parts in the denotation provided in Figure 2.

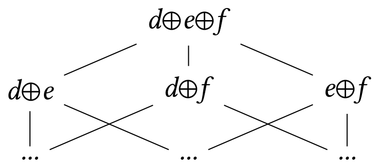


Figure 2: Denotation of mass nouns

Object-mass nouns in Asante Twi follow the same patterns concerning count nouns and quantity comparison as English object-mass nouns. Shown in (17), object-mass and count nouns can both combine with stubbornly distributive adjectives, but not mass nouns.

(17) a. **Count nouns**

n-kótérɛ keseɛ
PL-lizard big
'big lizards'

b. **Mass nouns**

*m-mógya keséɛ
NC-blood big
'big blood'

c. **Object-mass nouns**

á-kútúó keséɛ
NC-orange big
'big orange(s)'

Furthermore, object-mass terms like *nyadoa* 'garden eggs'[6] are compared on a cardinal scale. (18) is true even in a situation where Akosua's chairs outweigh Kofi's garden eggs, such as if Akosua had two very large chairs, but Kofi had only five garden eggs.

(18) Kofí wɔ n-yadoa bébréé sen Akósúá á ɔ-wɔ n-konyá.
Kofi has NC-garden.eggs more COMP Akosua REL she-has PL-chair
'Kofi has more garden eggs than Akosua has chairs.'

The same arguments made for an atomic denotation for object-mass nouns in English can also be made for Asante Twi. Therefore object-mass nouns in Asante Twi denote a full semilattice with atomic parts, the same as plural count nouns in both Asante Twi and English.

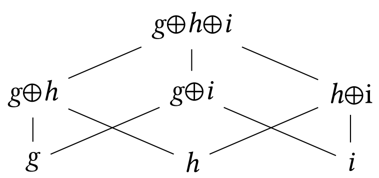


Figure 3: Denotation of object-mass nouns

3.2 Distribution

English's and Asante Twi's object-mass nouns greatly overlap in their distribution, as shown in the previous sections of this paper. The most notable exception is the ability to combine with numerals, which is what designates English as a Type I and Asante Twi as a Type III language. The second difference is simply due to Asante Twi lacking a set of quantifiers that can only apply to count nouns, like English's *many* or *fewer*. *Dódó* and *kakra* can combine with mass terms, but have no counterpart that only combines with count terms. The table below displays the similarities in distribution of object-mass nouns between the two languages.

[6] Garden eggs are a vegetable that is similar to eggplant, which commonly used in Ghanaian cuisine.

Table 3: English and Asante Twi object-mass nouns compared

	English	Asante Twi
Direct combination with numerals	X	✓
Morphological number distinction	X	X
Combination with mass quantifiers	✓	✓
Combination with count quantifiers	X	NA

3.3 Domain & Development

While similarities in the semantic denotation and syntactic distribution can be seen in English and Asante Twi's object-mass nouns, this class of nouns in each language has developed from different diachronic sources: English's from functional aggregates and Asante Twi's from the now-defunct noun class system. Interestingly, the different diachronic sources has resulted in the object-mass nouns in each language to appear in different lexical semantic domains.

3.3.1 English

English's object-mass nouns are said to be functional aggregates that emerged from their function or associated event (Grimm & Levin 2012). For example, nouns like *change* and *furniture* are related to their verbal counterparts *to change* (in this case money) and *to furnish*, respectively. Furthermore, *letters* can be considered *mail* through the event (or intention) of being transmitted through the postal system. Additionally, Chierchia (1998) argues that object-mass nouns, due to denoting collections, denote singleton properties, just as mass nouns do. He states that "[object-mass] nouns arise as a 'copy cat' effect", essentially mimicking the form of mass nouns due to having the same formal feature of denoting singleton properties (Chierchia 1998: p. 139).

These nouns also have their component parts as nearly synonymous counterparts, which receive count syntax shown in (19).

(19) a. Natalie has too much change in her pocket.
Natalie has too many coins in her pocket.

b. Kendra bought three pieces of furniture.
Kendra bought two chairs and a couch.

c. Sarah sent some mail.
   Sarah sent a few letters.

These component parts can be substituted for the object-mass noun in question without changing the core meaning of the sentence. However, the component parts may be more informative than the object-mass noun, which denotes a collection of individuals. All chairs are furniture, but not all furniture is chairs. Therefore, the object-mass nouns and their component parts are not necessarily interchangeable.

While object-mass nouns in English fall into the domain of functional aggregates that came about as a result of de-verbalization of their associated events and mimicry of mass nouns, object-mass nouns in Asante Twi differ from English greatly both in domain and diachronic development.

3.3.2 Asante Twi

As mentioned in Section §2.2, object-mass nouns in Asante Twi fall into the domains of agricultural produce, some animals, some artifacts, and things that tend to come in multiples or pairs. Additionally, mass and object-mass nouns exhibit similarities in the form of their noun class prefixes, both types of nouns being marked with [a-] or [N-], similarly to the plural count nouns. Osam (1994) discusses the concept of 'frozen plural nouns' where he lists many of the nouns now understood to be object-mass nouns. His explanation for why these nouns have a form similar to the plural count nouns in Akan is due to the fact that these nouns are often paired or tend to come in large numbers. Furthermore, in his discussion of the Akan noun class system, Osam (1994) links mass nouns and those lacking a morphological number distinction, stating, "the nasal prefix is used to mark liquids and some mass nouns...the nasal prefix is also found in certain words which do not have any plural distinction" (Osam 1994: pp.131-132). This link between object-mass and mass nouns in form and semantic domain could suggest that these were once part of the same noun class before the system had decayed to its current state. Further hints that this might be the case comes from comparing Asante Twi with other related languages and proto-language reconstructions. For example, going all the way back to Proto Niger-Congo, noun classes 5 and 6 are primarily characterized by containing masses and liquids, though these classes also include count nouns in the same domains in which the object-mass nouns in modern Asante Twi are found (Christaller 1881, Denny & Creider 1986, Maho 1999, Di Garbo et al. 2019, Good 2020 a.o.).

Other languages with observed object-mass nouns lack the type of noun class systems typical of Niger-Congo languages. With Asante Twi's object-mass nouns seeming to develop out of its former noun class system, this suggests that noun class system attrition would then be a newly-attested source for object-mass nouns. Since "the mass-count distinction seems to have a role similar to that of nominal features such as gender (± masculine) or animacy (± animate)," and "listing a potentially count noun as a singleton property is essentially a matter of lexical choice," I suggest that noun class is also be such a nominal feature which influences the choice to list a noun as denoting a singleton property, creating object-mass nouns (Bale & Gillon 2020: p.2); (Chierchia 1998: p.139). Retaining the nominal feature of a certain noun class, even as the system decayed is the most likely diachronic source of object-mass nouns Asante Twi.

Despite the differences in the lexical domains and diachronic sources of object-mass nouns in English and Asante Twi, Asante Twi's overall system of the mass-count distinction is much more similar to that seen in Type I languages than that seen in Type III languages. The following section explores the differences between Asante Twi and prototypical Type III languages.

4 Variation between Type III languages

Chierchia (2021)'s definition of Type III languages reads: "Numerals freely combine with any type of noun. In combination with cognitively count nouns (e.g., *three cats*), numerals have the meaning they do in English. In combination with mass nouns they have a 'container' or a 'quantity of' reading (not necessarily *'standard' quantity of*)" (p. 22). Following this definition, Asante Twi fits into Type III, same as Yudja, Nez Perce, and Indonesian. However, Asante Twi displays some features that set it apart from its Type III sisters.

4.1 Pluralization

Asante Twi exhibits a singular-plural distinction for most object nouns. For Yudja and Nez Perce, however, plural marking is restricted by animacy. Lima (2014) states that, "only [+human] nouns can be pluralized in Yudja" [p.542]. As for Nez Perce, plural marking can be seen on multiple elements of a phrase, though nominal plural marking is only available for human nouns. Plurality can be seen on the verb for animate nouns (including humans), and all nouns show plural marking on the adjective (Deal 2017). Nez Perce substance nouns can also have plural marking on their modifying adjective, shown in (20).

(20) 'ilexn̂i ti-ta'c tuutnin'
a.lot PL-good flour
'a lot of good portions of flour' Nez Perce (Deal 2017: ex. 55)

Asante Twi contrasts with Yudja and Nez Perce, as animacy doesn't play a role in number marking in Asante Twi, and any substance nouns often already have a very plural-like prefix of [a-] or [N-]. Furthermore, the plurality of the noun is not marked on the verb, the modifying adjective, or other elements of the phrase.

4.2 Combination with Quantifiers

Concerning quantifiers, Lima (2014) reports a count quantifier in Yudja, *itxïbï* 'many'. *Itxïbï* can combine with substance nouns, but has the interpretation of 'many portions of *x*'. However, Lima (2014) does not report a set of quantifiers that can only combine with either object or substance nouns. For Nez Perce, there is no object-substance distinction in which nouns can combine with which quantifiers. In fact, Deal (2017) states that "all quantifiers combine with all nouns" (p. 26). These languages both differ from Asante Twi, as Asante Twi has *dódó* 'much' and *kakra* 'a bit of', which are restricted to combining with only mass (substance) and object-mass nouns as was shown in (12) .

4.3 Count Adjectives

In both Yudja and Nez Perce, stubbornly distributive predicates can combine with all nouns. As shown in (21) and (22) for Yudja and Nez Perce, respectively, substance nouns combined with an adjective such as 'big' have an interpretation of 'big portion of *x*'.

(21) ma de urahu asa dju a'u
who big flour have
'Who has a big portion of flour?' Yudja (Lima 2014: ex. 31b)

(22) himeeq'is kuus
big water
'big portion of water' Nez Perce (Deal 2017: ex. 59a)

Substance nouns in Asante Twi, however, are not able to combine with count adjectives[7] There are rare exceptions, as in (23b), but these cases get an idiomatic reading, rather than a portion reading.

(23) a. *m-mógya keséɛ
NC-blood big
'big blood'

b. m-frámá keséɛ
NC-air big
'strong wind'

5 Accounting for numeral-substance combinations

In the above constructions, Asante Twi displays a grammatical mass-count distinction, while Nez Perce and Yudja do not. Though these three languages do converge on their ability to combine numerals and substance nouns, Asante Twi again stands out from the other Type III languages in how it accomplishes this construction.

5.1 Yudja & Nez Perce

To account not only for the numeral-substance combination in Nez Perce, but all of the phenomena listed in Section §4, Deal (2017) proposes that these languages still retain their cognitive distinction between object and substance nouns. She states that the denotations of some nouns are inherently quantized (anti-divisive; no proper part of p is also p (Krifka 1989)), such as *picpic* 'cat', and others are inherently homogeneous (a noun is both cumulative and divisive (Bunt 1985)), such as *kike't* 'blood'. For nouns with inherently homogeneous denotations, Deal (2017) puts forward a phonologically null element, α_n, shown in (24).

(24) $[\![\alpha_n]\!]^g = \lambda P \lambda x. AT_n(P)(x)$
where $AT_n = g(n) =$ *the* n^{th} *atomization function* (Deal 2017: ex. 64)

[7] Osam (1994), Osam (1993), as well as some native speakers have reported that Asante Twi's adjectives do in fact agree in number. However, Osam (1993) also notes that all adjectives have lost their prefix when modifying a singular noun and that only some adjectives take the plural marker. The majority of the consultants for this project did not display any agreement on the adjective, save for one who inconsistently displayed number agreement on the adjective. This could be indicative of a language change in progress.

This operator maps homogeneous denotations to quantized ones, allowing for substance nouns, i.e., ones with homogenized denotations, to be in grammatically count contexts such as getting a portion reading when combining with count quantifiers or count adjectives and the ability to combine directly with numerals. A phrase like *lepit kike't* 'two blood' would have the structure as follows in (25).

(25) [*lepti* 'two' [Num: - [α_n *kike't 'blood'*]]] Nez Perce (Deal 2017: ex. 66b)

Finally, Deal (2017) shows how adding her α_n-operator to Yudja constructions of substance nouns with numerals, count quantifiers, and count adjectives also makes correct predictions. For a more detailed discussion, see Deal (2017: pp. 36-41).

5.2 Asante Twi

While the α_n-operator could apply to substance nouns in Asante Twi to permit combination with numerals, it would make predictions for other constructions that aren't attested in the language. For example, Deal (2017) accounts for count adjectives being able to combine with substance nouns such as *himeeq'is kuus* 'big water' (26), through the α_n-operator creating an atomization of *kuus* 'water', to which *himeeq'is* 'big' can attach. The resulting phrase denotes being big and an element of a contextually relevant atomization of *water*.

(26) [*himeeq'is* 'big' [α_n *kuus 'water'*]] Nez Perce (Deal 2017: ex. 70b)
'big (glasses/puddles/buckets of) water'

Importantly, this is possible due to the conditions that Deal (2017) puts on α_n, where atoms instantiate the property of which they are an atomization and that atoms do not overlap with each other. The non-overlapping atoms are then able to combine with an adjective such as *himeeq'is* 'big', which can only have a distributive reading (Schwarzschild 2011). When applied to Asante Twi with the same conditions, the α_n would allow for constructions such as *nsuó keséɛ* 'big water', shown in in (27), to mean 'large, individuated protions of water'. Given that *nsuó keséɛ* is not acceptable in Asante Twi, applying the α_n-operator to substance nouns produces an incorrect prediction.

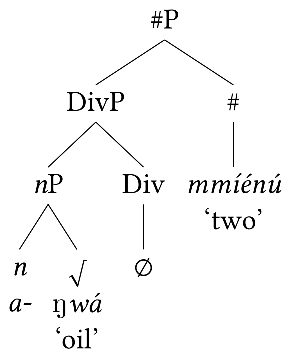
(27) *[[α_n *nsuó* 'water'] *keséɛ* 'big']
Intended: 'big (glasses/puddles/buckets of) water'

As an alternative proposal, I suggest that Asante Twi is making use of a null classifier to allow for substance nouns and numerals to freely combine while retaining a grammatical mass-count distinction for other constructions.

Asante Twi's numeral-noun constructions have the following structure: an *n*P containing a root and the former class marker, a DivP, which is responsible for division, and the #P, which makes the noun countable and hosts the numeral (Borer 2005). Beginning with the *n*P, Asante Twi's former noun class marker is on *n*, following Kramer's 2015 analysis of gender occurring on *n* cross-linguistically. I take noun class and gender to be accomplishing the same task: sorting nouns into two or more categories based on semantic, morphological, or arbitrary factors. The former noun class marker in Asante Twi is on *n*, along with its associated class and number features. As the noun class system has suffered attrition and seems to serve the function of marking number much more than that of a noun class marker, I suggest that the class feature, if present, is also on *n*. Such a class feature would be the feature of the former mass-denoting class as an example.

According to Borer (2005), a #P necessarily selects a DivP, as a noun must be divided in order to be countable, though it is possible for the Div head to be phonologically null. Given this structure, I suggest that Div, which is necessarily projected in numeral-noun constructions hosts a phonologically null classifier as shown in the structure in (28b).

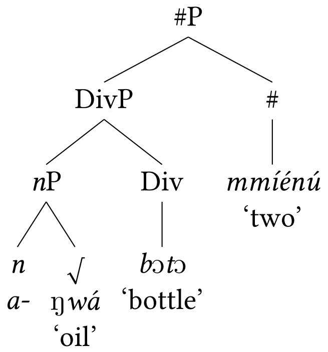
(28) a. aŋwá mmíénú
oil two
'two contextually salient containers or individual instances (e.g., two bottles or two drops) of oil '

b. 

While this classifier is often null, it is possible to use an explicit classifier or container phrases when counting portions of substance nouns. (28) then has the same structure as (29), but *bɔtɔ* 'bottle' does not appear in the PF of (28a).

(29) a. aŋwá bɔtɔ mmíénú.
oil bottle two
'two bottles of oil'

b.



Therefore, any time a substance noun is combined with a numeral, a classifier is necessary, whether pronounced or not. When null, the measurement or container is licensed by the context in which the utterance is spoken. For example, if a shop is selling oil in bottles, and no other container, *aŋwá mmíénú* could only mean 'two bottles of oil'. Conversely, if *aŋwá mmíénú* is uttered when some oil was spilled on the counter, it can only mean 'two drops of oil'. When the context does not provide a clear measure or container, a cooperative speaker would then choose to express the classifier overtly to avoid ambiguity, as in (29a).

6 Conclusion

At first glance, Asante Twi seems to be, according to Chierchia's 2021 mass-count typology, a Type III language, as it can freely combine numerals with any noun. Due to belonging to Type III, object-mass nouns should not be observable in Asante Twi. However, as discussed in §2, we can see clearly that object-mass nouns are attested in the language. With a closer look, Asante Twi's mass-count system is not so different from English in that both systems present a difference in how the grammar treats substance and object terms, aligning them with mass and count syntax, respectively. Asante Twi's ability to combine numerals and substance terms directly is simply a result of the availability for the classifier to be unpronounced, though it remains in the structure. Ultimately, rather than how a language treats numeral-noun combinations, it is the system of grammatical differentiation between mass and count that is a necessary precondition for a language to reanalyze some object nouns into formally mass nouns, resulting in object-mass nouns. The prototypically Type III languages, however, do not show a grammatical distinction in how they treat substance and object terms, only a cognitive one. In this way Asante Twi is more aligned with Type I languages than Type III languages.

Abbreviations

1 = first person, 2 = second person, 3 = third person, COMP = comparative, CL = classifier, DEF = definite, DEM = demonstrative NC = noun class, NEG = negative P = perfective, PST = past, PL = plural, REL = relativizer, REM.PAST = remote past, SG = singular, SUBJ = subject

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